Code for Good 2026 – Singapore

Challenge Statement

Organization Overview

Name: Austism Partnership

Website: https://autismpartnership.com.sg/en/

Mission: We believe that with quality Autism Partnership Method (APM) treatment, an intensive behavior treatment based on the principles of Applied Behavioral Analysis (ABA), individuals with Autism Spectrum Disorder are likely to achieve their fullest potential and achieve the greatest degree of independence and highest quality of life possible.

Autism Partnership, as the name suggests, believes that working and collaborating with families, schools, and everyone involved is fundamental to achieving a child's success.

Scale (beneficiary, geography & reach): Autism Partnership provides one-to-one therapy for children aged 6 and under its own method, the Autism Partnership Method (APM). It currently supports about 30–35 children, plus a couple overseas, with roughly one therapist per child. Each child has a team of therapists, and teams are reshuffled every year, so children learn to work with different people.

Problem Context & Current State

The organization provides individualized, one-to-one services that depend on coordinated staffing, scheduling, and continuity of care. Unexpected absences must be reported by 7:00 a.m., giving the operations team a short window to assess available resources and publish a revised schedule by 7:30 a.m.

Reassignment decisions must consider therapist availability, capability, familiarity with the child, service requirements, location, workload, and continuity. Current processes rely heavily on spreadsheets, messaging tools, supervisor input, and staff knowledge, while relevant information is fragmented across systems and not always structured for rapid decision-making.

Once a revised schedule is confirmed, affected therapists, supervisors, and families must be identified and notified promptly. The process also involves sensitive personal and service information that requires appropriate access controls, secure handling, and a clear record of decisions and changes.

Current operational reality

- **Resource allocation is complex:** Assignments must balance availability, capability, service requirements, and continuity.

- **Disruptions require rapid action:** Unexpected changes create time pressure and require multiple factors to be assessed quickly.
- **Information is fragmented:** Relevant data is spread across different tools and is not always structured, complete, or readily accessible.
- **Changes affect multiple parties:** Once a revised plan is confirmed, impacted individuals must be identified and informed quickly and consistently so that the change can be carried out.

Challenge Statement

How might we help Autism Partnership design a secure and auditable solution that supports rapid resource reallocation when operational disruptions occur, while maintaining service continuity, practical feasibility, and timely notification of impacted individuals.

The solution should:

1. **Support rapid replanning** when disruptions occur, while applying relevant operational and service constraints.
2. **Make decisions transparent** by recording the rules, trade-offs, approvals, and rationale behind changes.
3. **Enable coordinated execution** by identifying impacted individuals and issuing timely, role-appropriate notifications after scheduling decisions are made.
4. **Improve operational insight** through reliable reporting on disruptions, decisions, outcomes, and recurring constraints.

Stakeholder Universe

Primary users

- **Admin/Operation team:** Owns rapid reassignment and morning schedule publication; needs speed, accuracy, and confidence under time pressure.
- **Supervisors / Clinical leads:** Provide clinical oversight, advise on fit, coach sessions when required, and manage family alignment.

Secondary users

- **Therapists (Junior/Senior):** Need timely, clear assignments, including overlaps or shadowing, and minimal confusion about daily changes.
- **Parents / Families:** Need timely notification of changes, continuity wherever possible, and consistent progress updates.
- **Leadership / Program managers:** Need visibility into capacity constraints, disruption of drivers, and operational performance trends.

Data Landscape

Key Systems/Channels in use

- **Session and progress tracking system (HiRasmus)**
  - Holds session logs and progress visualizations.
  - Reports are generated in the system but manually extracted for sharing.
- **Scheduling artifacts such as spreadsheets and schedules (Excel)**
  - Used for day-to-day scheduling and historical review.
  - Often used to infer prior assignments from past schedules.
- **Messaging applications and group chats (WhatsApp)**
  - Used for urgent replacements, internal coordination, and stakeholder notifications.
  - Fast and familiar, but limited in structure, auditability, and retrieval.

Data Gaps

- Familiarity history is not maintained as a reliable, structured record.
- Expertise and fit indicators are not consistently captured in a format usable for scheduling.
- Matching rules and decision rationale are not formalized or consistently recorded.
- Decision and notification records are dispersed across channels, limiting traceability.

Key Constraints & Considerations

- **Time sensitivity:** The solution must support fast, reliable decisions within short operational windows.
- **Service quality and continuity:** Reallocations should preserve appropriate capability, continuity, and stakeholder expectations wherever possible.
- **Mandatory controls:** Non-negotiable policy, safeguarding, eligibility, and access requirements must be enforced consistently.
- **Operational feasibility:** Availability, timing, location, workload, and other practical dependencies must be considered when proposing changes.
- **Data limitations:** The solution should remain useful when information is incomplete or unstructured and should support lightweight improvement of data quality over time.
- **Change execution and auditability:** Once decisions are confirmed, affected individuals should be identified and notified consistently, with a clear record of the changes and rationale.
- **Privacy and security:** Sensitive information must be protected through appropriate access controls, secure handling, and accountable use.